CHENGZHI ZHANG

chengzhiz.github.io
TSRB 85 5th St NW,
Atlanta, GA 30332
czhang694@gatech.edu
Google Scholar

I am a **Human-centered** HCI researcher with a background in interaction design, AI, and computer science. My research interests lie at the intersection of **AI and User Experience** and AI literacy. Recently, I design and implement public-facing interaction installations to enhance **AI literacy**, with a particular focus on **Large Generative Models**.

EDUCATION

2023-2027 **Georgia Institute of Technology** | Atlanta, GA
Ph.D. in Digital Media, Minor in AI, Expressive Machinery Lab
Advisor: Brian Magerko **Expected graduation: 2027**

2021–2023 **Carnegie Mellon University** | Pittsburgh, PA
M.S. in Computational Design
Dissertation: *Generative Image AI for Design*
Committee: Daragh Byrne, Nikolas Martelaro, Paul Pangaro

2016-2020 **Hunan University** | Changsha, China
B.ENG. Design: Industrial Design
with one year exchange in Politecnico di Milano, Italy
Thesis Project: *Optia-Smart Glasses for Myopia Control and Prevention*

EMPLOYMENT

2023–2027 **Graduate Research Assistant**
Expressive Machinery Lab, Georgia Institute of Technology, Atlanta
- Led UX research for AI literacy exhibits deployed at major museums (Griffin Museum of Science and Industry and others), engaging 300+ visitors.
- Employed mixed-methods evaluation: contextual interviews (150+), surveys, and interaction data logging to assess educational impact and usability.
- Translated research insights into iterative design improvements, directly increasing exhibit clarity and learning outcomes.
- Managed full research life cycle: literature review, prototype design, on-site data collection, analysis, and reporting.

2023 **Graduate Research Assistant**
Carnegie Mellon University, Pittsburgh
- Led a mixed-methods UX study with 11 architecture students, using design tasks, observation, and interviews to assess an AI sketch-to-image workflow.
- Defined actionable design requirements for future AI tools, such as integrated sketching platforms and domain-specific model tuning.
- First-authored paper presented at ACM Creativity and Cognition (C&C '23).

2019 **UI/UX Design Intern**
Alibaba Group, Hangzhou, China
- Conducted immersive field research in transit and pharmacy settings to observe user interactions, pain points, and workflows.
- Delivered actionable insights that directly shaped the product roadmap and feature prioritization for Alipay services.
- Enabled data-driven design decisions by connecting observed behaviors to specific product enhancements.

PUBLICATIONS

Pre-print Articles

Chengzhi Zhang, Brian Magerko. "Generative AI Literacy: A Comprehensive Framework for Literacy and Responsible Use". In: *arXiv preprint arXiv:2504.19038* (2025).

Yuwen Lu, Yuewen Yang, Qinyi Zhao, **Chengzhi Zhang**, Toby Jia-Jun Li. "AI Assistance for UX: A Literature Review Through Human-Centered AI". In: *arXiv preprint arXiv:2402.06089 and in submission* (2024).

Conference Papers

Hasti Darabipourshiraz, Sophie Rollins, Milka Trajkova, **Chengzhi Zhang**, Yasmine Belghith, Tom McKlin, Jessica Roberts, Brian Magerko, Duri Long. "Designing and Evaluating Museum Exhibit Prototypes to Foster Middle Schoolers' AI Literacy through Creativity and Embodiment". In: *Proceedings of the 20th International Conference on Tangible, Embedded and Embodied Interaction*. 2026.

Workshops Papers (Refereed)

Yuwen Lu, Ziang Tong, Qinyi Zhao, **Chengzhi Zhang**, Jia-Jun Li Toby. "UI Layout Generation with LLMs Guided by UI Grammar". In: *ICML 2023 Workshop on AI and HCI*. 2023.

Demos, Videos, and Work-In-Progress (Refereed)

Chengzhi Zhang. “Designing for Public Enlightenment: Enhancing Generative AI Literacy on Socio-technical Aspects in Informal Learning Spaces”. In: *Proceedings of the 20th International Conference on Tangible, Embedded and Embodied Interaction*. 2026.

Chengzhi Zhang, Dashiel Carrera, Daksh Kapoor, Jasmine Kaur, Jisu Kim, Brian Magerko. “Sound Clouds: Exploring ambient intelligence in public spaces to elicit deep human experience of awe, wonder, and beauty”. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems Creative AI Track: Humanity*. 2025. URL: https://openreview.net/forum?id=rOk1B3TYbj.

Chengzhi Zhang, Chelsi Alise Cocking, Milka Trajkova, Zoe Lacy Mock, Gemma Tate, Cassandra Naomi Monden, Brian Magerko. “Fostering AI Literacy with LuminAI through Embodiment and Creativity in Informal Learning Spaces”. In: *Proceedings of the 16th Conference on Creativity & Cognition*. 2024, pp. 476–481.

Chengzhi Zhang, Weijie Wang, Paul Pangaro, Nikolas Martelaro, Daragh Byrne. “Generative Image AI Using Design Sketches as input: Opportunities and Challenges”. In: *Proceedings of the 15th Conference on Creativity & Cognition*. 2023, pp. 254–261.

Yuwen Lu, **Chengzhi Zhang**, Iris Zheng, Toby Jia-Jun Li. “Bridging the Gap between UX Practitioners’ work practices and AI-enabled design support tools”. In: *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 2022, pp. 1–7.

Patents

Yingying Zheng, Cuijun Zheng, **Chengzhi Zhang**. “A bagged platelet oscillation device”. ZL 2021 2 0038780.4. 2021.

HONORS & AWARDS

2023 **MSCD Research Support Fund ($1000)** School of Architecture, CMU

2022 **George W. Anderson, Jr. Award ($1000)** School of Architecture, CMU

2022 **MSCD Research Support Fund ($1000)** School of Architecture, CMU

2020 **First Prize Scholarship** Hunan University

2018 **Third Prize Scholarship** Hunan University

RELEVANT COURSEWORK

2026 **PSYC-6020 Statistical Analysis II** Prof. James Roberts, GT

2024 **CS-4731 Game AI** Prof. Mark Riedl, GT

2023 **11-485 Deep Learning** Prof. Bhiksha Raj, CMU

2022 **10-601 Machine Learning** Prof. Henry Chai, CMU

2022 **05-630 Programming Usable Interface** Prof. Scott Hudson, CMU

2022 **48-758 Responsive Mobile Environments** Prof. Daragh Byrne, CMU

2021 **48-675 Design for Internet of Things** Prof. Daragh Byrne, CMU

PROFESSIONAL SERVICE

Reviewing Service

2023 ACM CHI Late Breaking Work
2024 ACM CHI Late Breaking Work, CSCW
2025 ACM CHI, DIS, IDC
2026 ACM CHI, DIS, IDC

Academic Service

2026 **Graduate Students Advisory Board** GT Ivan Allen College of Liberal Arts

Student Volunteer

2024 ACM CHI

SKILLS & CERTIFICATES

CERTIFICATES

2022 **Mathematics for Machine Learning: Linear Algebra** Awarded through Coursera

SKILLS

UX Research User Interview, Contextual Inquiry, Survey Design, Usability Evaluation, Thematic Analysis

Data Analysis R, SPSS, JMP

Design Adobe Creative Suite, UI/UX Design, Service Design

Programming Python, Front-end Development, C++

Machine Learning Pytorch, TensorFlow, Scikit Learn